Assessing the Role of Data Complexity in Addressing Challenges in Learning from Imbalanced Data

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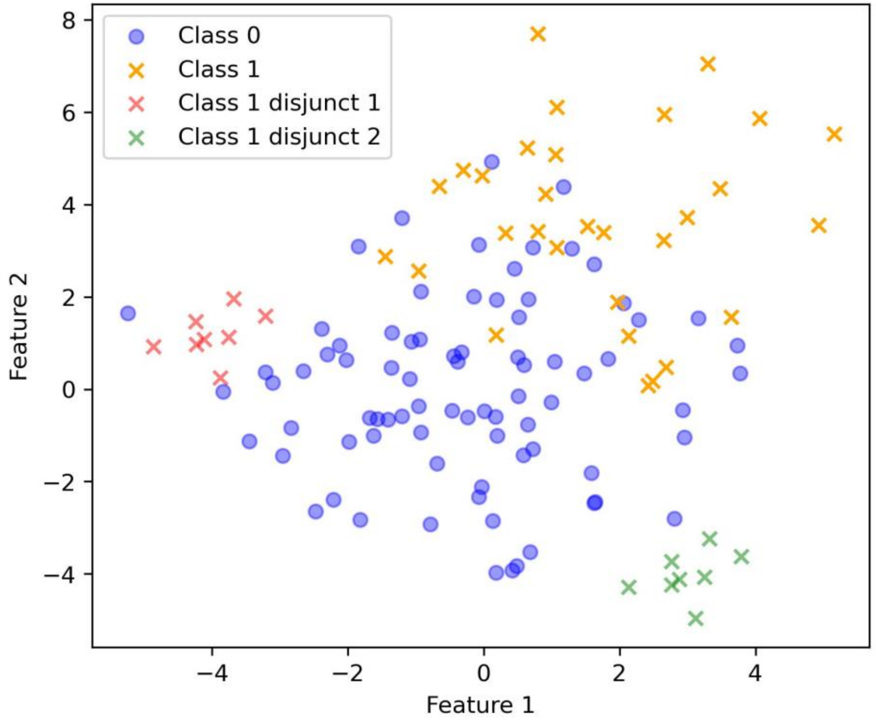
Symposium on Data Science and Statistics 2025, Salt Lake City, Utah

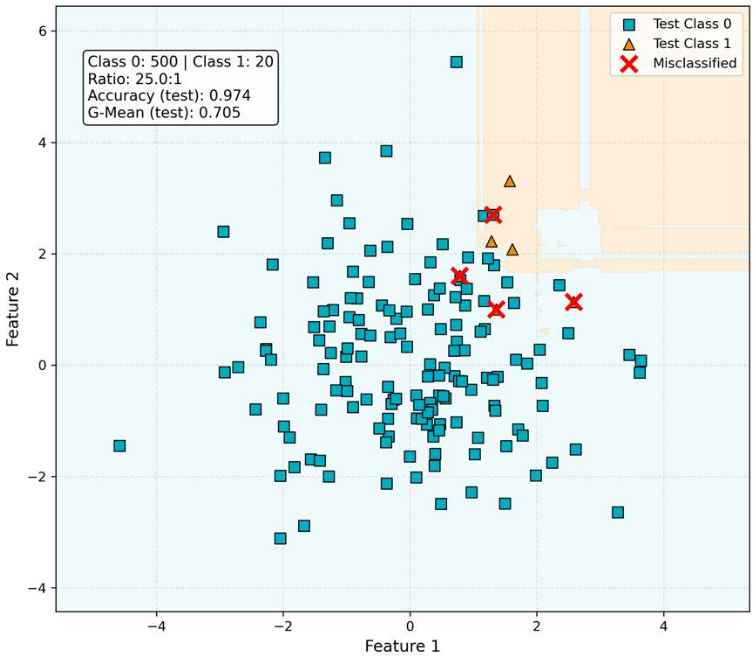
MOTIVATION

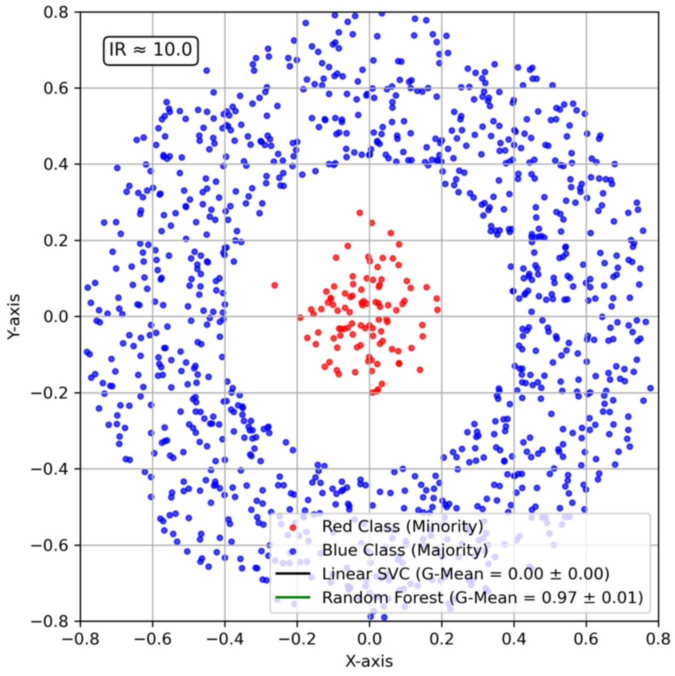
- Imbalanced data ≠ Only challenge
- Dimensionality, Separability, Small Disjuncts, Sparsity, Nonlinearity



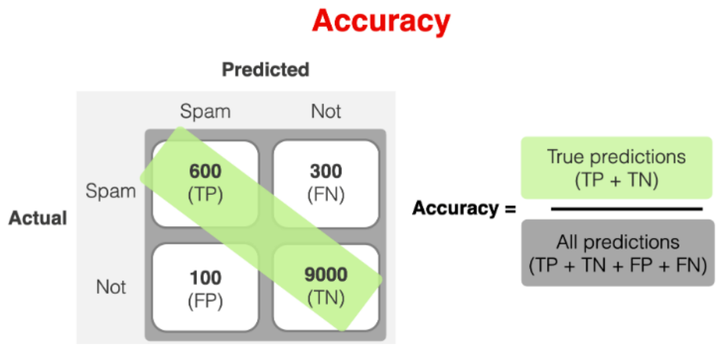
RESEARCH QUESTIONS

- How does data complexity impact model performance beyond class imbalance?
- How can we quantify different aspects of data complexity?
- Can complexity profiling guide data preprocessing and model selection?





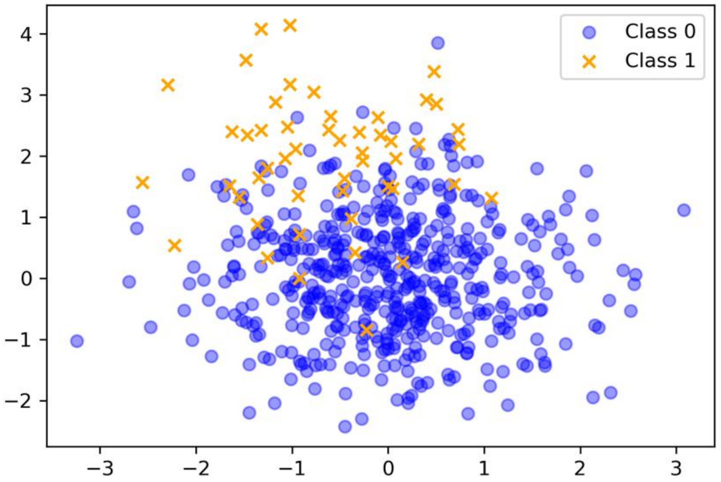
$$G_{mean} = \sqrt{\frac{TP}{TP+FN} \times \frac{TN}{TN+FP}} = \sqrt{Sensitivity \times Specificity}$$

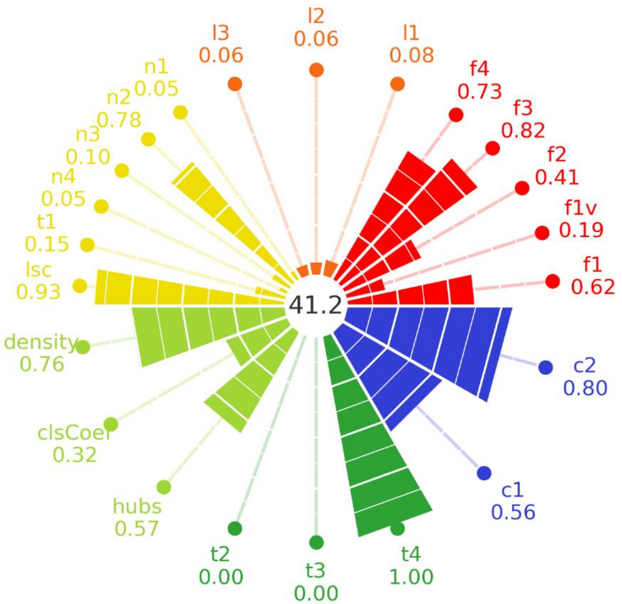


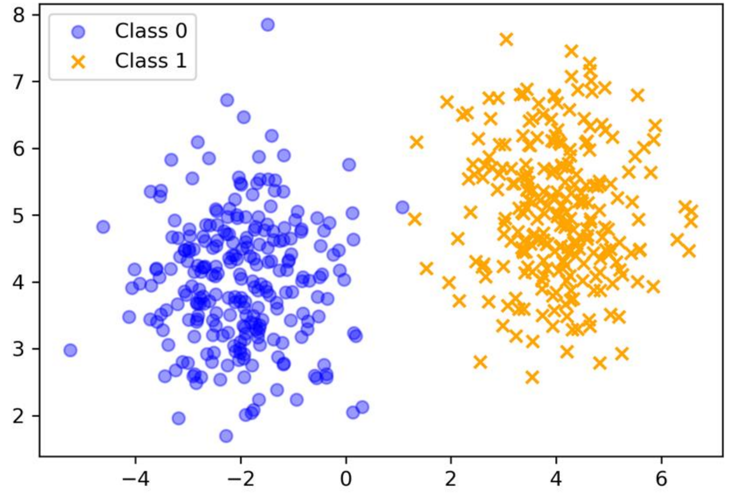
METHODS

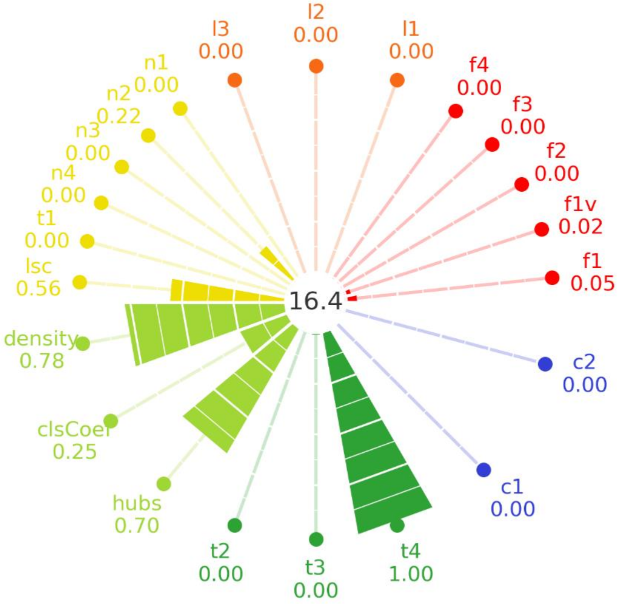
- **Feature-based:** Discriminative power of individual and combined features
- **Linearity:** Degree of linear separability
- **Neighborhood:** Local class density and cohesion
- **Network:** Graph-based structural information
- **Dimensionality:** Sample-to-feature sparsity
- **Imbalance:** Unequal distribution of class labels

METHODS

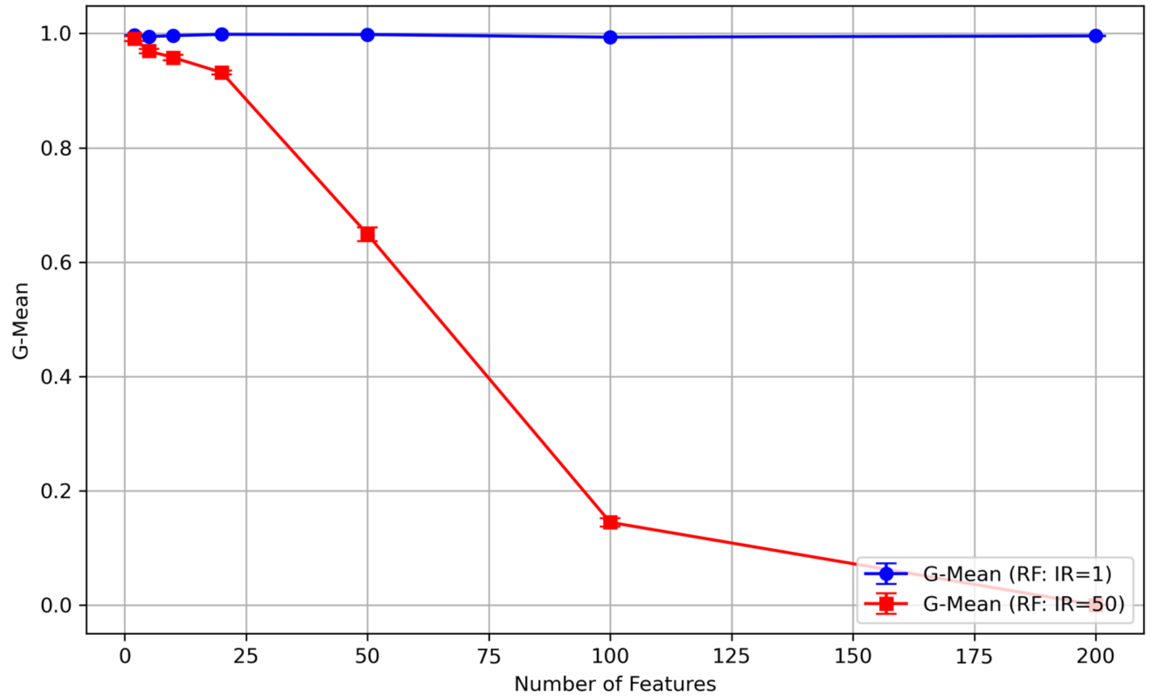

Class 0
Class 1
4
3
2
1
0
−1
−2
−3
−2
−1
0
1
2
3


l2
0.06
l3
0.06
l1
0.08
n1
0.05
f4
0.73
n2
0.78
f3
0.82
n3
0.10
f2
0.41
n4
0.05
f1v
0.19
t1
0.15
f1
0.62
lsc
0.93
41.2
density
0.76
c2
0.80
clsCoef
0.32
c1
0.56
hubs
0.57
t2
0.00
t3
0.00
t4
1.00


Class 0
Class 1
8
7
6
5
4
3
2
−4
−2
0
2
4
6

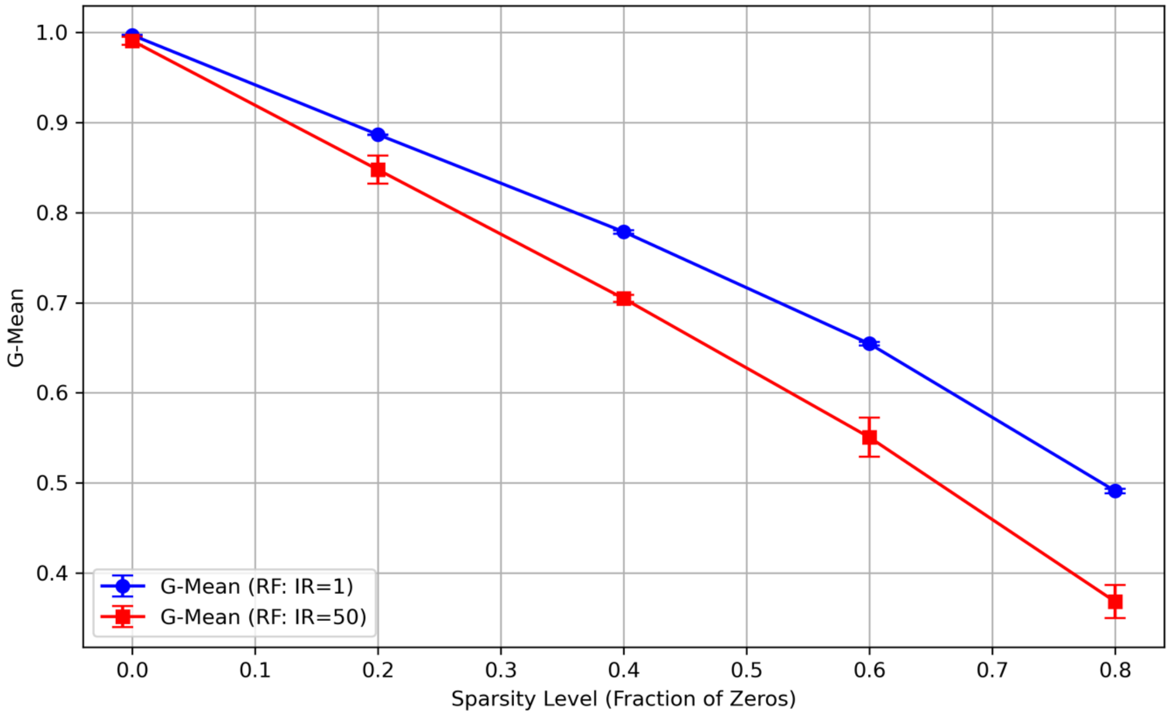

l2
0.00
l3
0.00
l1
0.00
n1
0.00
f4
0.00
n2
0.22
f3
0.00
n3
0.00
f2
0.00
n4
0.00
f1v
0.02
t1
0.00
f1
0.05
lsc
0.56
16.4
density
0.78
c2
0.00
clsCoef
0.25
c1
0.00
hubs
0.70
t2
0.00
t3
0.00
t4
1.00

INFLUENCE OF COMPLEXITY



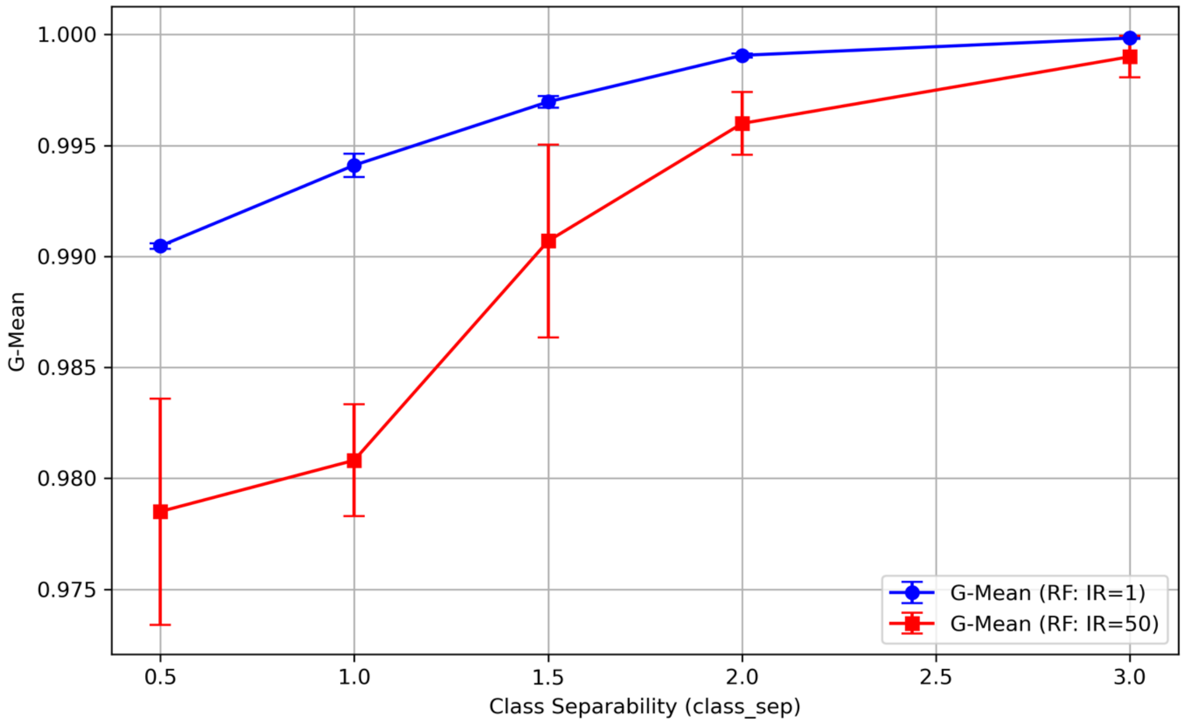
- G-Mean declines with more features under imbalance

INFLUENCE OF COMPLEXITY



- High sparsity reduces model performance

INFLUENCE OF COMPLEXITY

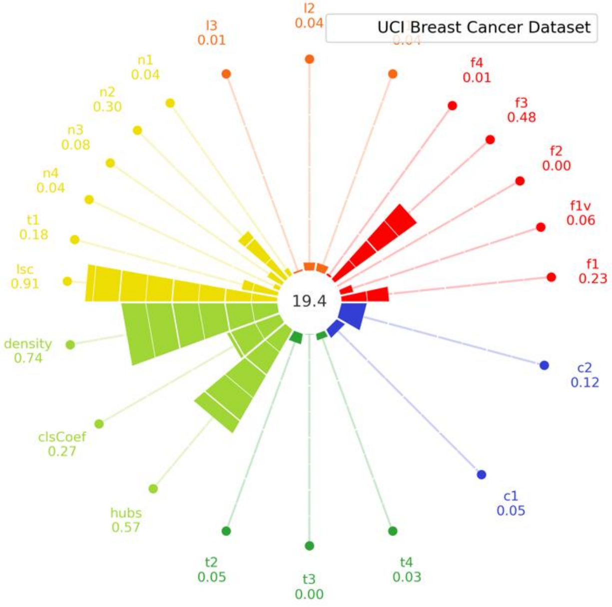


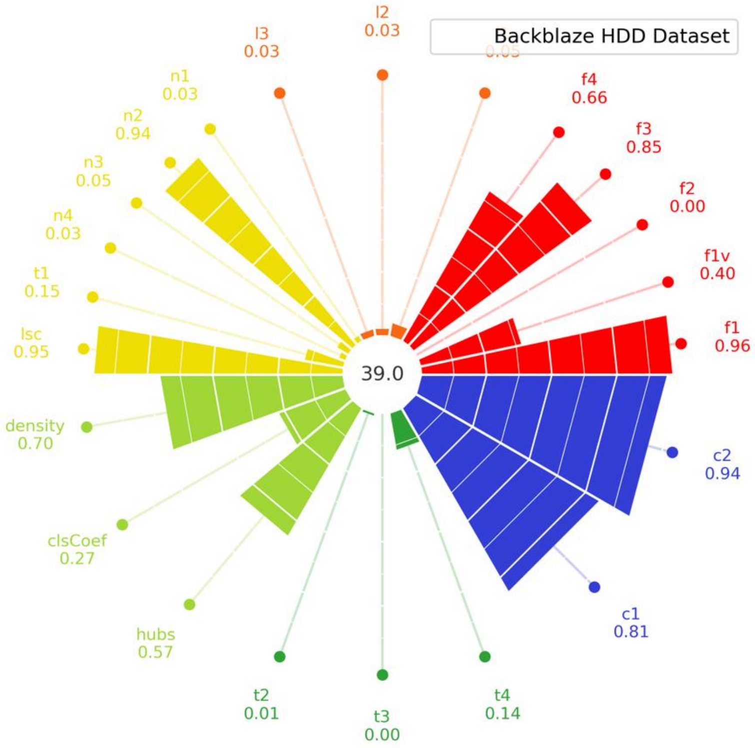
- Higher class separability improves performance

RESULTS

Datasets:

Dataset	# Instances	IR
HDD (high complexity)	2,487,576	33.91
UCI (low complexity)	569	1.68

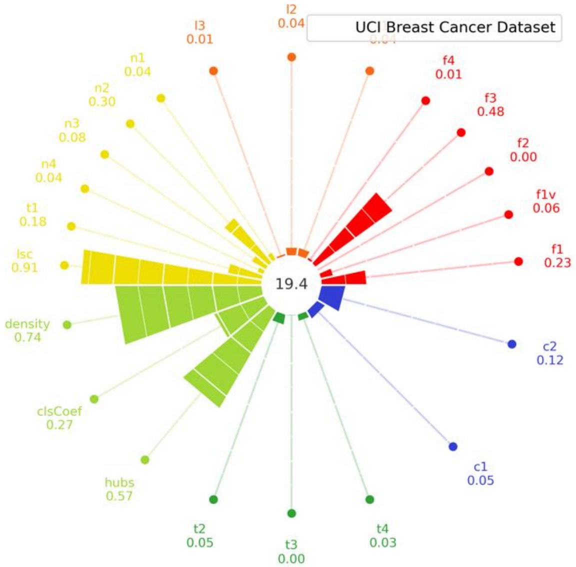


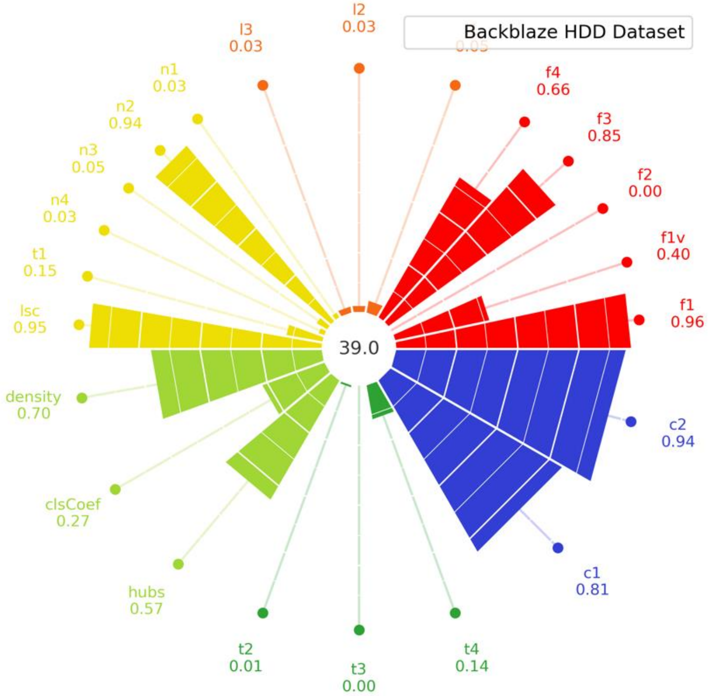


RESULTS

Why This Dataset Is Hard ?

- High Feature Overlap (F1 = 0.96, F3 = 0.85, F4 = 0.66)
- Local class boundaries are unclear (N2 = 0.94, LSC = 0.95)
- Overconnected regions increase overlapping (Density = 0.70, Hubs = 0.57)
- Severe Imbalance (C1 = 0.81, C2 = 0.94)





RESULTS

Dataset	Metric	Without Class Weight	With Class Weight
UCI	G-mean	0.9258 +/- 0.0020	0.9307 +/- 0.0045
	AUC	0.9260 +/- 0.0021	0.9307 +/- 0.0045
HDD	G-mean	0.0000 +/- 0.0000	0.6070 +/- 0.0017
	AUC	0.5000 +/- 0.0000	0.6650 +/- 0.0009

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2. Komorniczak, J., & Ksieniewicz, P. (2022). problexity--an open-source Python library for binary classification problem complexity assessment. arXiv preprint arXiv:2207.06709.
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